

Homework 3 – Distributed Systems

Monsoon 2025

IIIT Hyderabad

Instructions:

- Due date: October 6, 2025, 10 PM
 - Each questions carries 4 points.
 - Any case of copying will result in zero points and possibly more penalties.
 - Submit either a typed copy or scanned handwritten sheets, in PDF format via moodle.
1. Read about the way Unix (and Linux) file systems use multiple indirect inode blocks to store files. Find the maximum size a file can have on such a system by using suitable parameters. Why does not GFS use such a hierarchical mechanism? What are some other advantages/disadvantages of these two approaches: flat vs hierarchical storage models for metadata?
 2. In the algorithm of Lamport for mutual exclusion, show a scenario where a processor/site P satisfies condition OTHERS-OK but not condition TOP. Similarly, show a scenario where a process/site P satisfies condition TOP but not condition OTHERS-OK. Use four or more processors to make the example non-trivial.
 3. Recall the algorithm to handle crash failures in distributed consensus. Show an example of the algorithm working with two faulty nodes out of a total of 6 nodes. How many rounds are needed in your example in various cases?
 4. Recall the Phase King algorithm. Explain with an example as to why the algorithm fails when $n < 4f$.